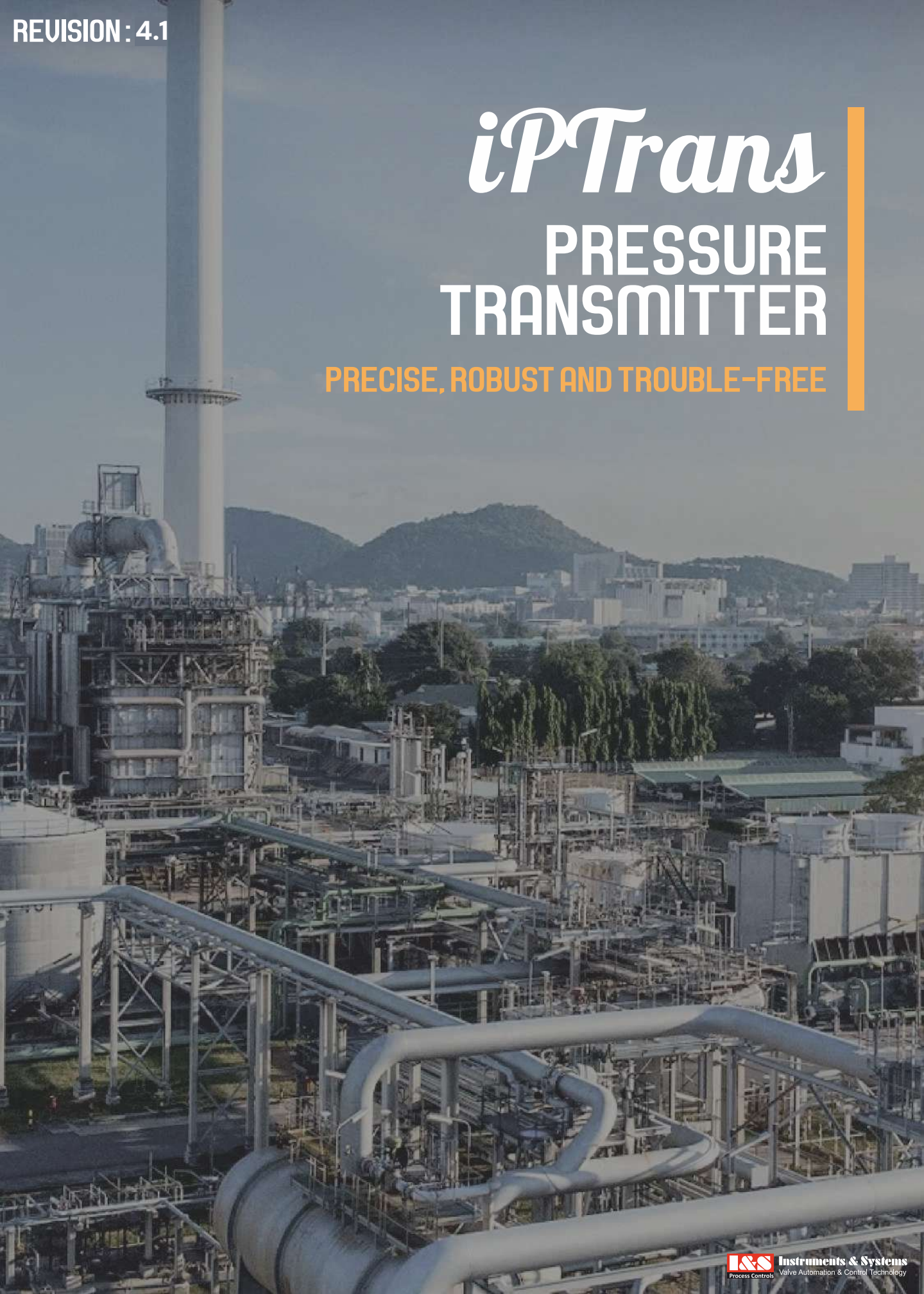
REVISION : 4.1

iPTrans

PRESSURE TRANSMITTER

PRECISE, ROBUST AND TROUBLE-FREE



I&S Process Controls
Instruments & Systems
Valve Automation & Control Technology

Why *iPTrans?*

In today's world where industry is moving towards industry 4.0, smart solution is required for high level of automation.

iPTrans pressure transmitter avoid unplanned downtime and improve product quality which in turn reduces project costs and start up time. The *iPTrans* Pressure Transmitters deliver total value across the entire plant life cycle, from construction to operations to maintenance.

As each industry has its own specific demands, whether they are requirements for safety-related devices or harsh, rugged environmental conditions. The *iPTrans* Pressure Transmitters has complete range of portfolio to meet industry applications.

Uncompromising precision

iPTrans pressure transmitters are a complete family of measuring instruments for measuring gauge, differential, and absolute pressure that includes a range of performance values, load capacities, and materials. *iPTrans* pressure transmitters has high measuring accuracy and ruggedness, the modular system features superb operating.



iPTrans

WITH REMOTE SEALS

The ***iPTrans***-200 is a Remote Seal that allows the pressure transmitter to make measures in situations where the immediate contact of the transmitter diaphragm with the process fluid it is not allowed. The remote seal is made up by one connection with the diaphragm and one capillary connection with filling fluid. The filling fluid transfers the pressure measured in the process fluid to the transmitter sensor. The capillary connection links the seal diaphragm to the transmitter.

The remote seals available in the ***iPTrans*** series are : Flanged "T" Type, Threaded, Pancake both models having an optional flush connection, Sanitary, Flanged with Extension and Pancake with Extension.

The ***iPTrans*** DP type is assembled with gauge and differential pressure transmitters. When used in food applications the connections are sanitary type. Level models are also available

The typical applications of the remote seal on the transmitter are:

- For corrosive process fluid
- For viscous process fluid or with solids in suspension
- The process fluid may freeze, crystallize or solidify
- Processes that require easy cleaning and others.





SMARTLINE MANIFOLDS SAFE ASSEMBLIES

SmartLine Accessories combine our market-leading pressure transmitters with quality manifolds in a single, integrated solution.

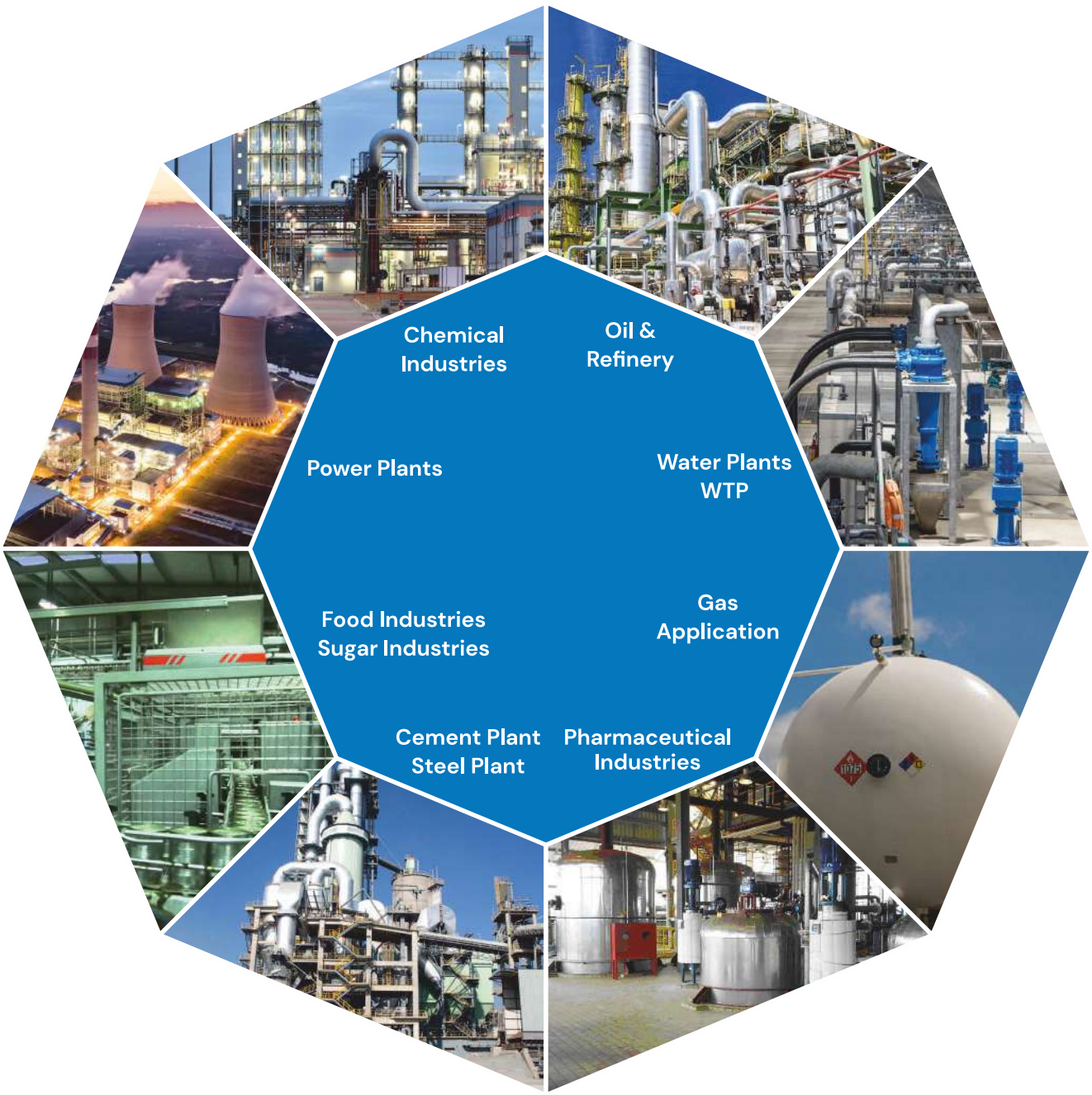
- Block and Bleed models for in-line pressure transmitters
- Two, three and five-way configurations
- Wide and compact body styles
- Traditional or wafer-style variants
- NACE compliant material options
- Available with factory integration and mounting brackets





I&S Process Controls Instruments & Systems Valve Automation & Control Technology

APPLICATION



TECHNICAL DATA

TYPE	*iPTrans* ISDPT-200	*iPTrans* ISPT-200
	 Side entry for DPT including oval flanges (Process Connection)	 Bottom entry for PT (Process Connection)
Service	Liquid, Oil, Gas or Vapor	Liquid, Oil, Gas or Vapor
Pressure Types	Differential Pressure	Absolute Pressure & Gauge Pressure
Working Principle	Capacitive, Inductive, Piezoresistive & Silicone Resonant	Capacitive, Inductive, Piezoresistive & Silicone Resonant
Max. Turndown Ratio	100:1	100:1
Accuracy	±0.05% of Span	±0.05% of Span
Long-Term Stability	±0.1%	±0.1%
Static Pressure Effect	±0.05% FS	±0.05% FS
Ambient Temp. Effect	±0.05% of Span	±0.05% of Span
Communication	4...20 mA/HART	4...20 mA/HART
Power Supply	9 to 45 V DC	9 to 45 V DC
Storage Temp. Limit	-40 to +85° C	-40 to +85° C
Ambient Temperature	-20° C ~ +70° C (with LCD)	-20° C ~ +70° C (with LCD)
Operating Temperature with Capillary	Upto 400° C	Upto 400° C
Measured Medium Temperature	-40 to +125° C	-40 to +125° C
Max. Pressure	See Table	1.5 times of FS
Diaphragm Materials	Stainless Steel, Hastelloy, Tantalum, Monel, Gold	Stainless Steel, Hastelloy, Tantalum, Monel, Gold
Response Time	<90mS*	<90mS*
Protection Class	IP66 / IP68 (Standard)	IP66 / IP68 (Standard)
Certifications/Approvals	ATEx, PESO & SIL2 Approved	ATEx, PESO & SIL2 Approved
Display	5 Digits programmable & 0-100% Bargraph 360° rotation LCD Display, Process Value, Output in mA	5 Digits programmable & 0-100% Bargraph 360° rotation LCD Display, Process Value, Output in mA
Zero/Span Configuration	Built in LCD with push button type with configurable engg. unit	Built in LCD with push button type with configurable engg. unit
High-Voltage Isolation	YES	YES
Multiple Field Adjustment	Function (Linear/Square Root), Zero	Function (Linear/Square Root), Zero
Functions	Trim, Lower Sensor Trim and Upper Sensor Trim	Trim, Lower Sensor Trim and Upper Sensor Trim
Installation	Easy to install and reliable with unique design	Easy to install and reliable with unique design
Zero & Span Drift	±0.01% at max. span	±0.01% at max. span

TECHNICAL DATA

TYPE	*iPTrans* ISDPT-200 	*iPTrans* ISPT-200 
Elevation and suppression by 100% of span	Elevation and suppression throughout the span	Elevation and suppression throughout the span
Damping	Adjustable	Adjustable
Dead Time	<40 ms	<40 ms
Update Rate	Reading Cycle Acrylic : 3/s	Reading Cycle Acrylic : 3/s
Turn On Time	<90ms	<90ms
Power Supply Effect	<0.0005% of URL/1 V	<0.0005% of URL/1 V
Vibration Effect	60 to 2000 Hz : 3 g in all 3 axes	60 to 2000 Hz : 5 g in all 3 axes
Mounting Position Effect	<-4 mbar (-0.06 psi)	<-4 mbar (-0.06 psi)
Failure Alarm	Self diagnostic feature available	Self diagnostic feature available
Self Diagnostic	Loop integrity diagnostic continuously monitors the 4-20 mA output signal	Loop integrity diagnostic continuously monitors the 4-20 mA output signal
Cover Orings	PTFE Seal, EP851	PTFE Seal, EP851
Electrical Connection	3 Pin Screwed, Plug & Socket Connector, Harting Han Connector, 1/2" NPT F	3 Pin Screwed, Plug & Socket Connector, Harting Han Connector, 1/2" NPT F
Vent & Drain	Built in Vent & Drain Plugs	Built in Vent & Drain Plugs
Humidity	0 to 100% RH	0 to 100% RH
Wetted Parts	SS316, SS316L, Hastelloy C276	SS316, SS316L, Hastelloy C276
Output Load Resistance	600 ohm load at 24 V DC supply	600 ohm load at 24 V DC supply
Reverse Polarity	Available	Available
Calibration Certificate	5 Point Calibration	5 Point Calibration
Life Span	Minimum 10 Year	Minimum 10 Year
Warranty	1 Year	1 Year

MODEL CODES

1. Model Number

Code	Description
IS-PT-200	Smart Pressure Transmitter / Remote Seal Pressure Transmitter
IS-DPT-200	Differential Pressure Transmitter / Remote Seal Differential / Flow / Level Transmitter

2. Pressure Type

Code	Description
A	Absolute Pressure Transmitter
G	Gauge Pressure Transmitter
D	Differential Pressure Transmitter

3. Pressure Range

Code	Differential Pressure Transmitter (D)	Gauge Pressure Transmitter (G)	Absolute Pressure Transmitter (A)
S1	-1.5 to 1.5 kPa -6 to 6 inH2O (-15 to 15 mbar)	-1 to 1 Bar G	0 to 1 Bar A
S2	-7.5 to 7.5 kPa -30 to 30 inH2O (-75 to 75 mbar)	-1 to 10 Bar G	0 to 5 Bar A
S3	-37.4 to 37.4 kPa -150.29 to 150.29 inH2O (-374 to 374 mbar)	-1 to 25 Bar G	0 to 10 Bar A
S4	-186 to 186 kPa -747.5 to 747.5 inH2O (-1.86 to 1.86 bar)	-1 to 50 Bar G	0 to 50 Bar A
S5	-690 to 690 kPa -2772 to 2772 inH2O (-6.9 to 6.9 Bar)	-1 to 100 Bar G	0 to 100 Bar A
S6	-2068.43 to 2068.43 kPa -300 to 300 psi (-20.68 to 20.68 Bar)	-1 to 400 Bar G	
S7	-6890 to 6890 kPa 1000 to 1000 psi (-68.9 to 68.9 Bar)	0 to 600 Bar G	
S8	-20680 to 20680 kPa 3000 to 3000 psi (-206.79 to 206.79 Bar)	0 to 1000 Bar G	

4. Static Pressure

Code/Pressure Range	
SP1	**150% of FS**
SP2	450 kg/cm²

MODEL CODES

5. Burst/Over Pressure

Code/Pressure Range	
SP3	**200% of FS**
SP4	>500 kg/cm²

6. Transmitter Output

Code	Description
A1	4-20mA with digital based HART protocol
A2	4-20mA without HART protocol

7. Process Connection

Code	Description
C1	1/2 inch NPT (Male)
C2	1/4 inch NPT (Male)
C3	1/2 inch NPT (Female)
C4	1/4 inch NPT (Female)
C5	1/2 inch BSP (Male)
C6	M20 x 1.5 (Male)
C7	3/4 inch NPT (Male)
C8	1/4 inch BSP (Male)
C9	1/4 inch BSP (Female)
C10	3/8 inch NPT (Male)
C11	3/4 inch NPT (Male) - Flushed
C12	1 inch NPT (Male)
C13	3/4 inch BSP (Male)
C14	1/2 inch BSP (Female)
C15	Flanged type - Direct Mount
C16	Remote seal with capillary

8. Isolating Diaphragm

Code	Description
D1	304 L Stainless Steel
D2	316 Stainless Steel
D3	316 L Stainless Steel
D4	Hastelloy C276
D5	Tantalum
D6	Monel
D7	Gold Plated
D8	SS310
D9	Duplex X220S

MODEL CODES

9. Sensor Fill Fluid

Code	Description
F1	Silicone

10. Housing Material

Code	Description	Conduit Entry Size
H1	Aluminum	1/2 inch NPT
H2	Aluminum	M20 x 1.5
H3	316 Stainless Steel	1/2 inch NPT
H4	316 Stainless Steel	M20 x 1.5
H5	304 Stainless Steel	1/2 inch NPT
H6	304 Stainless Steel	M20 x 1.5

11. Electrical Connection

Code	Description
E1	PVC blind plug on both sides
E2	SS cable gland on one side & other side PVC blind plug
E3	Brass cable gland on one side & other side PVC blind plug
E4	Plug and socket connector on one side & other side PVC blind plug
E5	Harting han connector on one side & other side PVC blind plug
E6	SS blind plug on one side & other side PVC blind plug
E7	SS blind plug on both sides
E8	SS cable gland on both sides
E9	Brass cable gland on both sides
E10	Plug and socket connector on both sides
E11	Harting han connector on both sides
E12	SS cable gland on one side & other side SS blind plug
E13	Brass cable gland on one side & other side SS blind plug
E14	Plug and socket connector on one side & other side SS blind plug
E15	Harting han connector on side & other side SS blind plug
E16	Brass cable gland on one side & other side Brass blind plug

12. Number of Seals

Code	Description
RS1	Remote Seal on single side or High Side
RS2	Remote Seal on both side (Both High & Low)
RS3	Remote Seal on low side

MODEL CODES

13. Secondary Fill Fluid (Capillary & Seal)

Code	Description
SF1	Silicone Oil DC 200
SF2	Silicone Oil 704
SF3	Fluorinate Oil CTFE
SF4	Silicone Oil M50

14. Flange Size

Code	Description
FS1	1 inch
FS2	1.5 inch
FS3	2 inch
FS4	3 inch
FS5	4 inch
FS6	6 inch
FS7	3/4 inch

15. Flange/Adaptor MOC

Code	Description
M1	304 L Stainless Steel
M2	316 Stainless Steel
M3	316 L Stainless Steel
M4	Hastelloy C276
M5	Tantalum
M6	Monel
M7	Gold Plated

16. Flange Extension Length

Code	Description
EL1	None
EL2	50.8 MM
EL3	76.2 MM
EL4	101.6 MM
EL6	152.4 MM
EL7	205 MM
EL8	254 MM
EL9	304.8 MM
EL10	350 MM

MODEL CODES

13. Secondary Fill Fluid (Capillary & Seal)

Code	Description
SF1	Silicone Oil DC 200
SF2	Silicone Oil 704
SF3	Fluorinate Oil CTFE
SF4	Silicon Oil M50

14. Flange Size

Code	Description
FS1	1 inch
FS2	1.5 inch
FS3	2 inch
FS4	3 inch
FS5	4 inch
FS6	6 inch
FS7	3/4 inch

15. Flange/Adaptor MOC

Code	Description
M1	304 L Stainless Steel
M2	316 Stainless Steel
M3	316 L Stainless Steel
M4	Hastelloy C276
M5	Tantalum
M6	Monel
M7	Gold Plated

16. Flange Extension Length

Code	Description
EL1	None
EL2	50.8 MM
EL3	76.2 MM
EL4	101.6 MM
EL6	152.4 MM
EL7	205 MM
EL8	254 MM
EL9	304.8 MM
EL10	350 MM

MODEL CODES

22. Manifold Valves

Code	Description	
MF24	2 way manifolds SS 304 – 1/2 inch NPT F	
MF34	3 way manifolds SS 304 – 1/2 inch NPT F	
MF54	5 way manifolds SS 304 – 1/2 inch NPT F	
MF26	2 way manifolds SS 316 – 1/2 inch NPT F	2 way manifolds SS 316L – 1/2 inch NPT F
MF36	3 way manifolds SS 316 – 1/2 inch NPT F	3 way manifolds SS 316L – 1/2 inch NPT F
MF56	5 way manifolds SS 316 – 1/2 inch NPT F	5 way manifolds SS 316L – 1/2 inch NPT F
MF86	None	

23. Product Certification

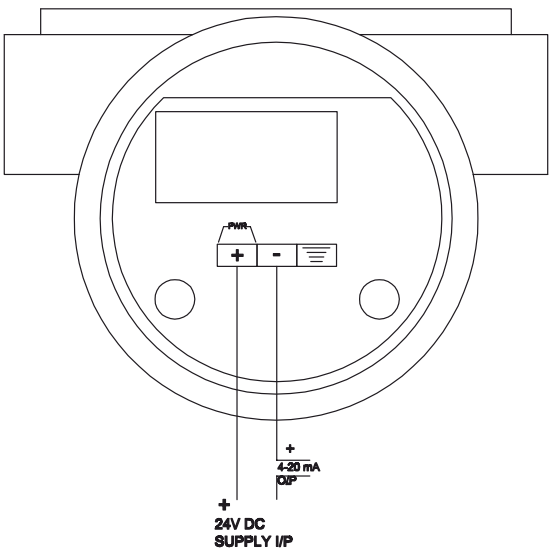
Code	Description
PC1	ATEx with one year warranty
PC2	Standard with one year warranty
PC3	SIL2 with one year warranty
PC4	IEC/IECEx Certificate
PC5	ATEx/SIL2 with three year warranty



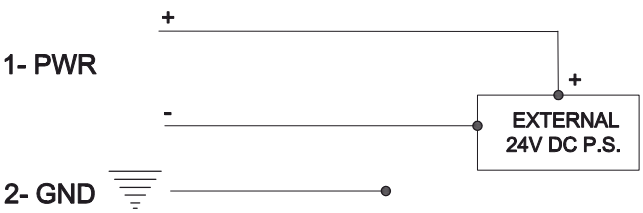
WIRING DIAGRAM

(a) Standard Electrical Connection

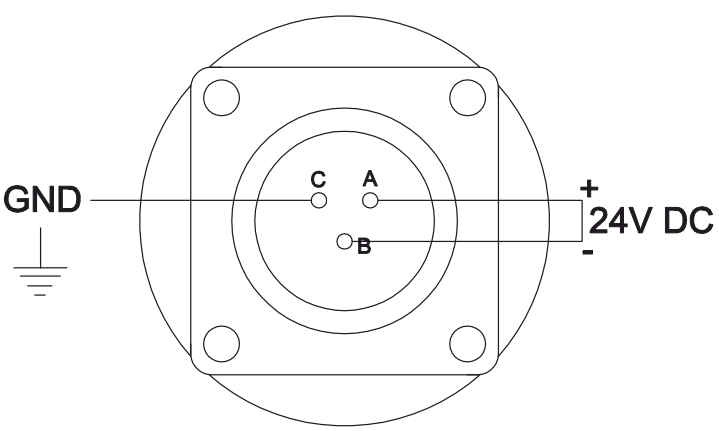
FIELD TERINATION VIEW :-



LOOP DIAGRAM :-

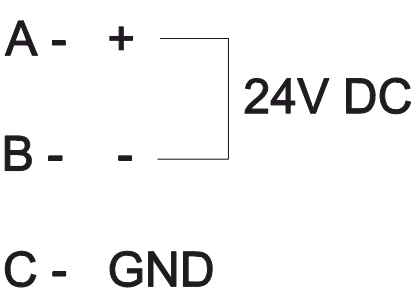


(b) Plug & Socket Type (3 Pin)



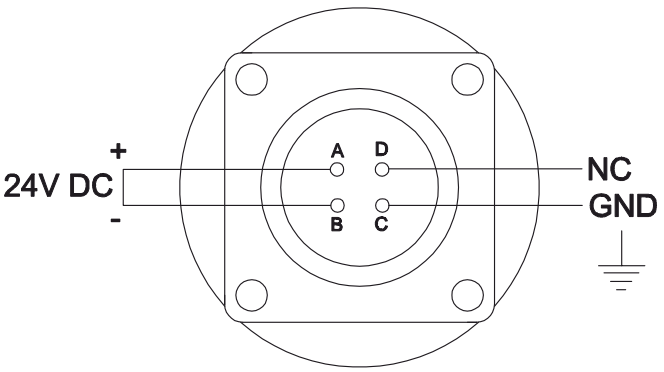
MALE PART

LOOP DIAGRAM:-



WIRING DIAGRAM

(c) Plug & Socket Type (4 Pin)



MALE PART

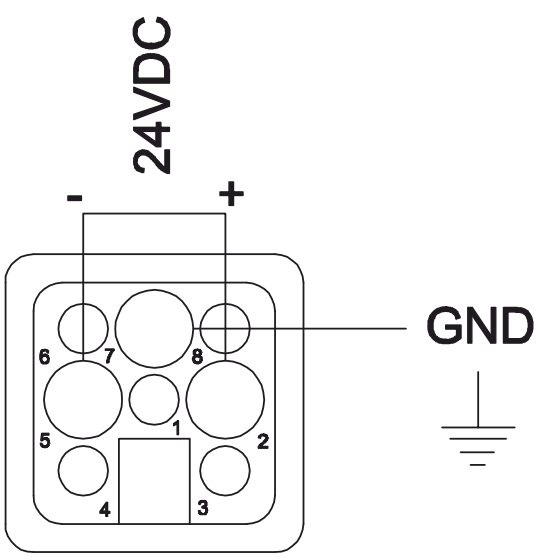
LOOP DIAGRAM:-

A - +
B - - 24V DC

C - GND

D- NC (NO CONNECTION)

(d) Harting Han Connector



LOOP DIAGRAM:-

2. +
5. - 24 VDC

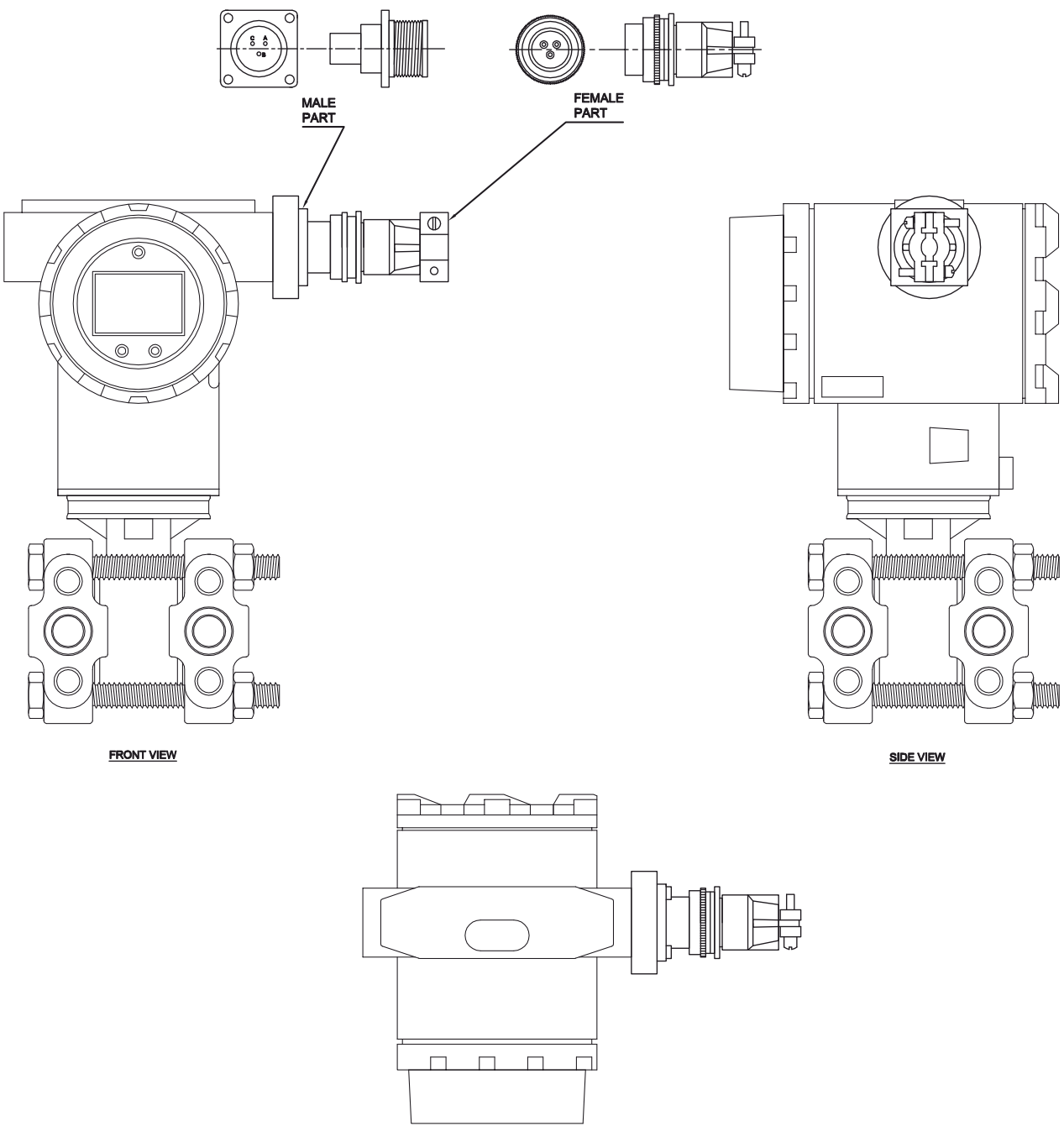
7. GND

ELECTRICAL / PROCESS CONNECTION

PLUG & SOCKET / DRAIN & VENT PLUG

iPTrans

ISDPT-200

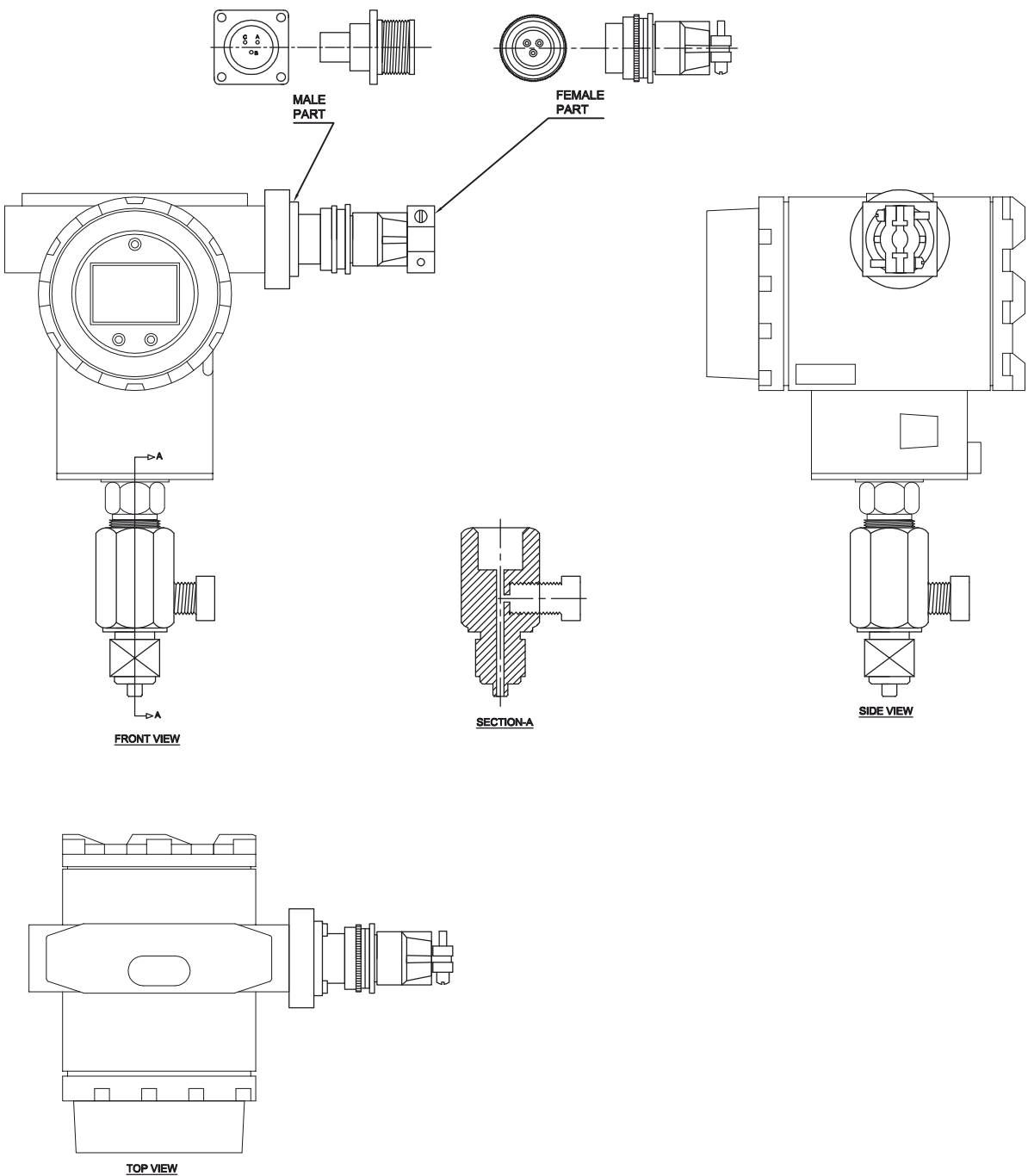


ELECTRICAL / PROCESS CONNECTION

PLUG & SOCKET / DRAIN & VENT PLUG

iPTrans

ISPT-200



iPTrans
ISDPT-200

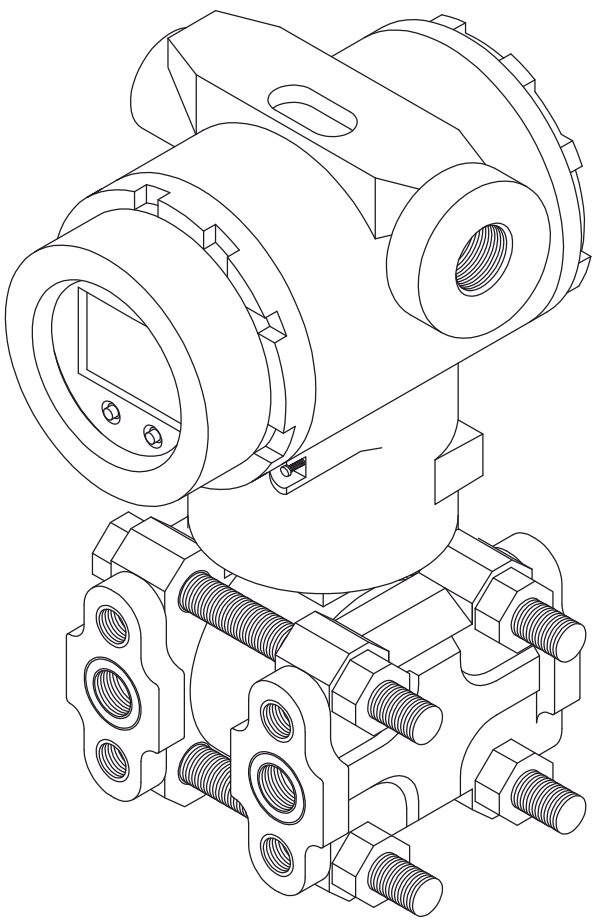


FIG.- ISOMETRIC VIEW

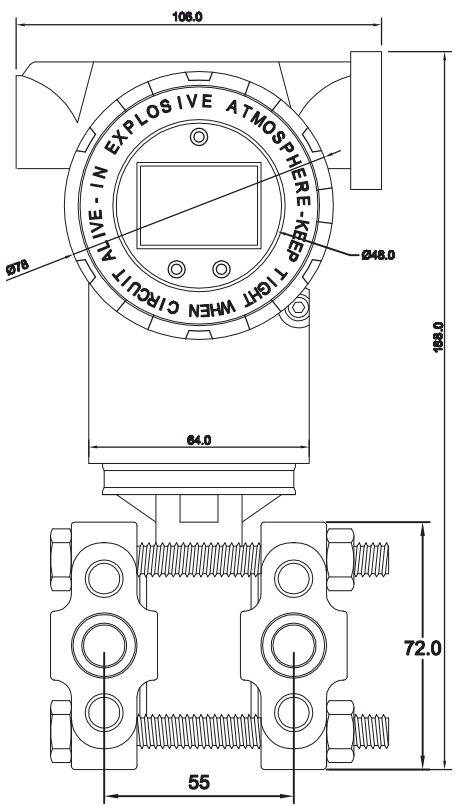


FIG.- FRONT VIEW

ALL DIMENSIONS ARE IN MM

DIAMENSIONS

iPTrans
ISPT-200

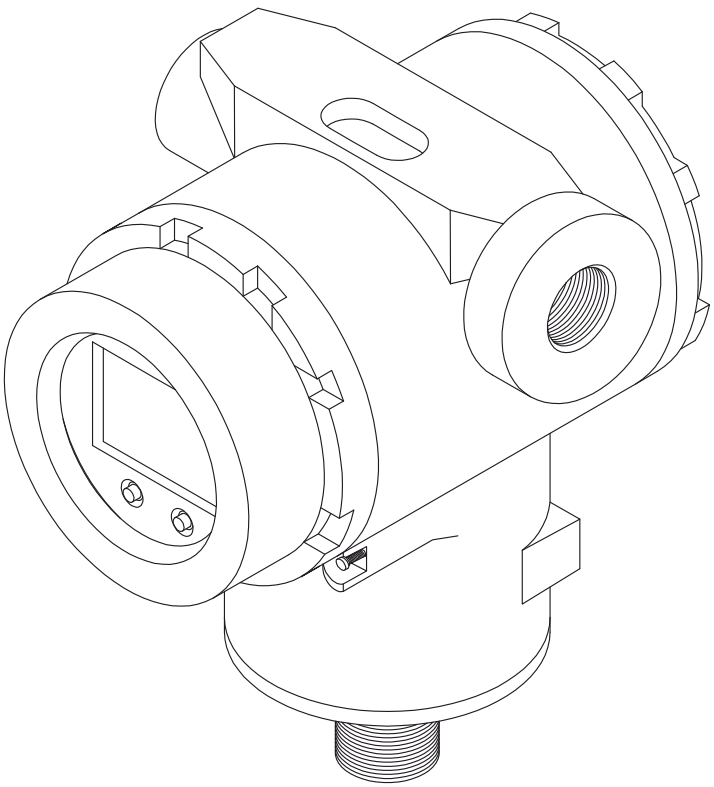


FIG.- ISOMETRIC VIEW

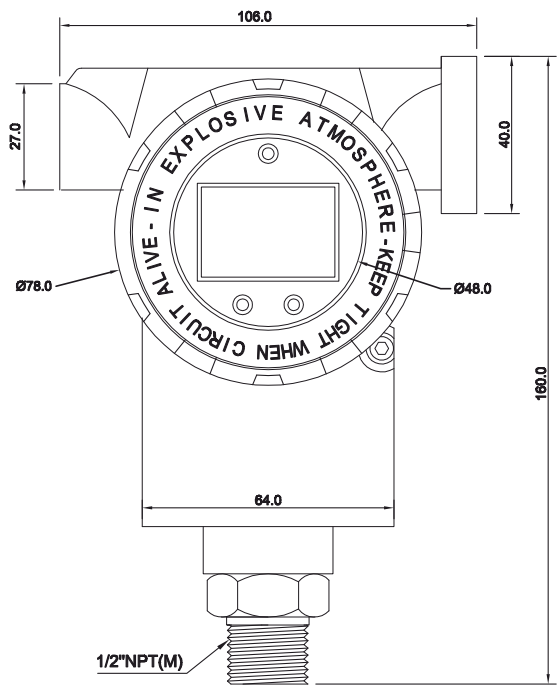


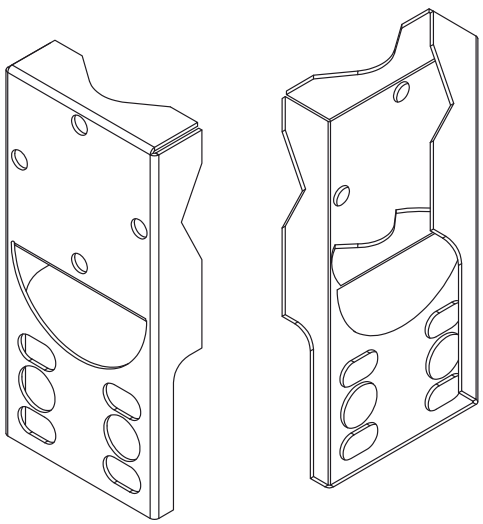
FIG.- FRONT VIEW

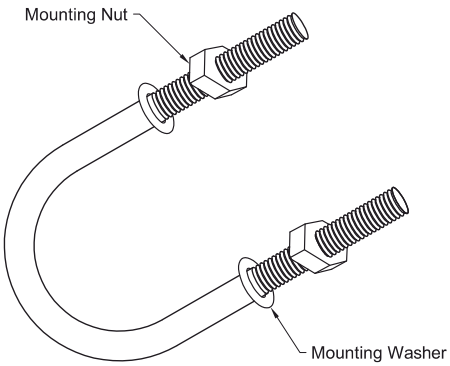
ALL DIMENSIONS ARE IN MM

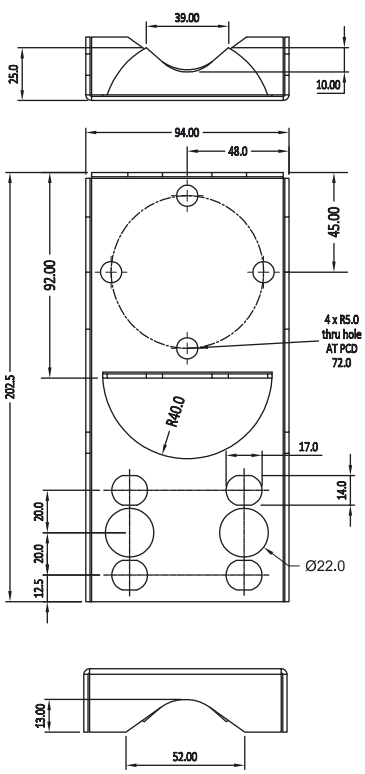
MOUNTING BRACKET DIAMENSIONS

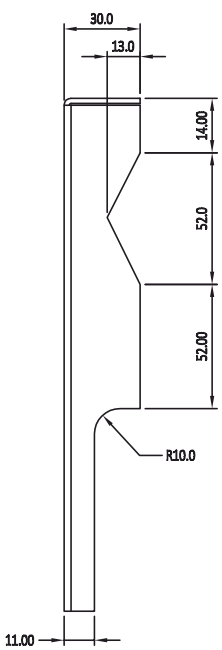
iPTrans
ISDPT-200

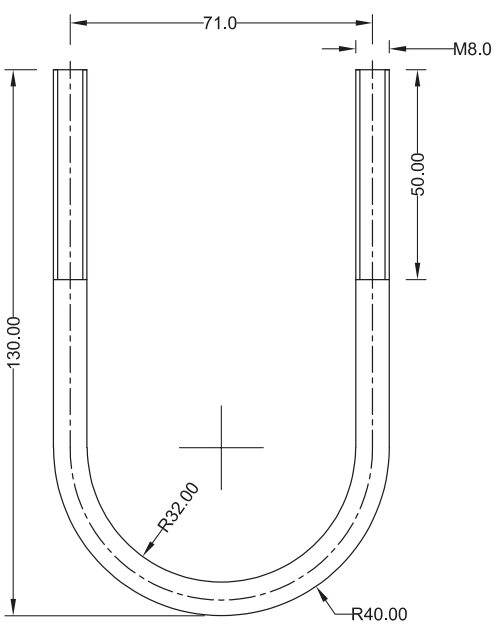
ISOMETRIC FIGURE







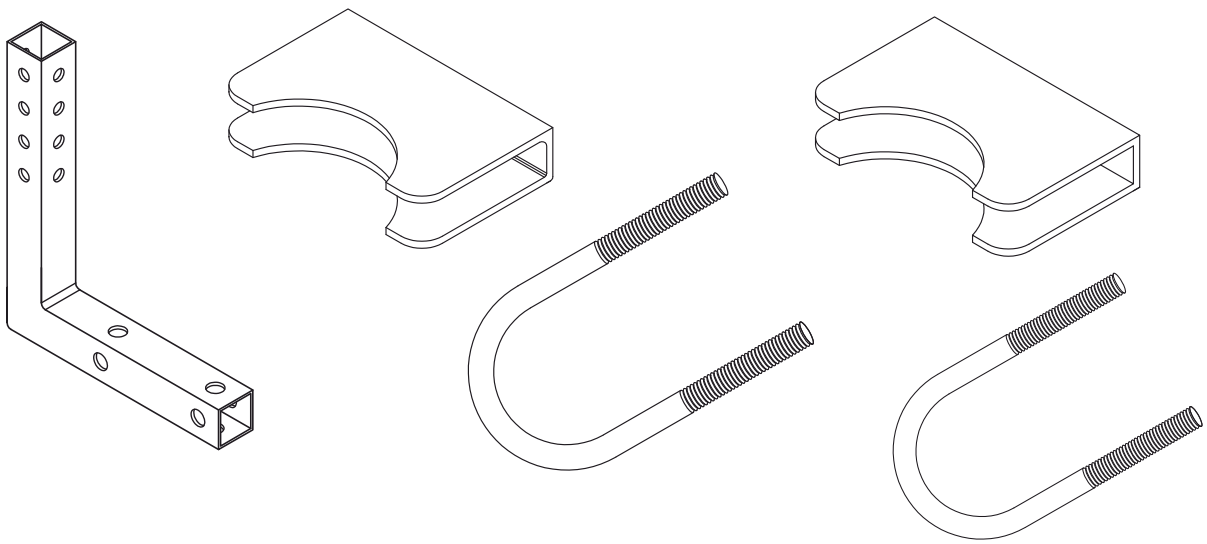


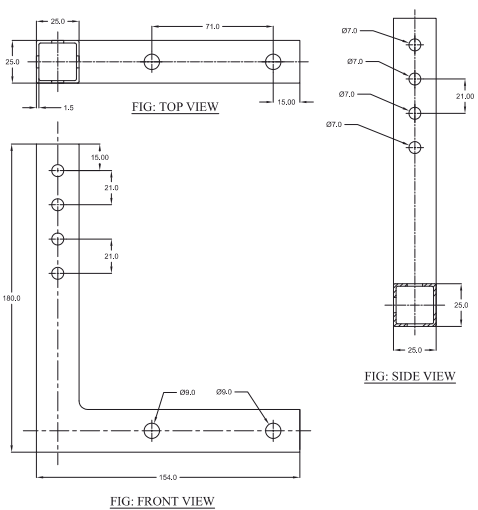


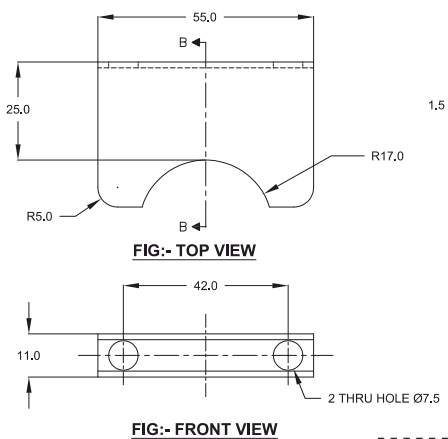
MOUNTING BRACKET DIAMENSIONS

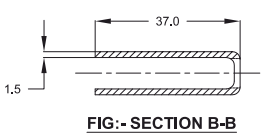
iPTrans

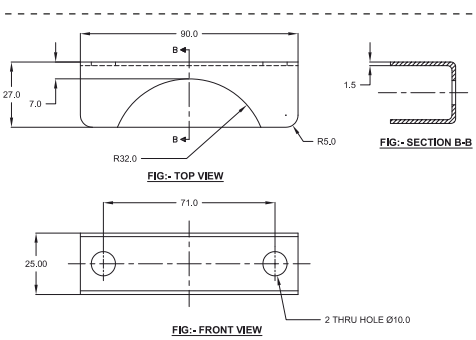
ISPT-200

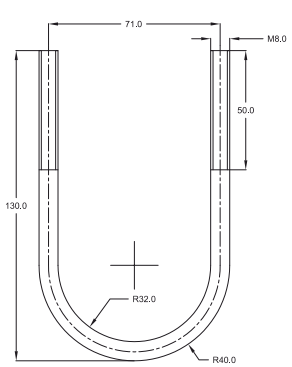


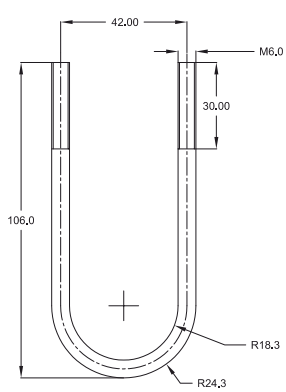


























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Valve Automation & Control Technology

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